

Alejandro Cruz Osorio

Instituto de Astronomía, Universidad Nacional Autónoma de México

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Nationality: Mexican

Date of birth: April 17 of 1984

Research Interests

Relativistic Astrophysics: Extragalactic Relativistic Jets – Shadows – Multiwavelength Emission – Accretion onto Compact Objects – General Relativistic Magnetohydrodynamics. Non-thermal Particle Acceleration.

General Relativity: Gravitational waves – Neutron and Quark Stars – Mixed stars.

Experience

Investigador Asociado C Aug.2023–Jul.2026

Instituto de Astronomía, Universidad Nacional Autónoma de México, México

Postdoctoral Fellow in Relativistic Astrophysics

Institut für Theoretische Physik, Goethe Universität Frankfurt, Germany. Mentor: Luciano Rezzolla, 2019–2023.

Postdoctoral Fellow in Numerical Relativity

2017–2019.

Departament d'Astronomia i Astrofísica, Universitat de València, Spain. Mentor: José A. Font Roda.

Postdoctoral Fellow in Relativistic Astrophysics

2016–2017

Frankfurt Institute for Advanced Studies, Frankfurt an Main, Germany. Mentor: Luciano Rezzolla

Postdoctoral Fellow in Astronomy

2015–2016

Instituto de Astronomía, UNAM. Mentor: F. Sánchez.

Full time interim professor

2014–2015.

Dep. of Computational Systems, Instituto Tecnológico Superior de los Ríos, México

Education

Ph.D. in Physics

2010-2014

Institute of Physics and Mathematics, UMSNH, México

Numerical implementation of the relativistic magnetohydrodynamics

M.Sc. in Physics

2008-2010

Institute of Physics and Mathematics, UMSNH, México

Numerical solution of the wave equation in Minkowski and Schwarzschild space-times in a domain containing the future infinite null

B.Sc in Physics-Mathematics

2002-2008

Faculty of Physical-Mathematical Sciences, UMSNH, México

Tracking null radial geodesics of spherically symmetric space-times.

Languages

Nahuatl: Native

Spanish: Native

English: Full professional proficiency

German: Basic

Awards and Recognitions

PRIDE-B: Programa de Estímulo por Equivalencia, UNAM, México. 2023–2028.

EHT Early Career Award 2022: For the outstanding contribution of Theoretical Modeling of Sagittarius A*.

The Event Horizon Telescope collaboration. Dec. 2022.

Wissenschaftspreis der Frankfurter Physik 2022 (Science Award of Frankfurt Physics 2022): Department of Physics Goethe University Frankfurt and Walter Greiner Gesellschaft zur Förderung der physikalischen Grundlagenforschung, Frankfurt, Germany. Jul. 2022.

SNII-1: National Researchers System Recognition by SECIHTI ¹, México. First period: Jan 2016 – Dec 2018. Second period: Jan 2019 – Dec 2022. Third period: Jan 2023 – 2027.

Research Grants

2025–2026 Project IA103725: Estudio de la polarización de la luz de sombras de agujeros negros y jets relativistas, Programa de Apoyo a Proyectos de Investigación e Innovación Tecnológica (PAPIIT)-DGAPA

2024–2026 Project CBF2023-2024-1102: Sombras de Agujeros Negros y Jets Relativistas como laboratorios para probar teorías de la gravedad, propiedades del plasma en discos de acreción y la aceleración de partículas, Ciencia Básica y de Frontera 2023-2024 (CONAHCYT)

2022–2026 Marie Curie Horizon Europe Staff Exchange action 2022 (NewFunFiCO) ²: Fundamental Fields and Compact Objects: new opportunities, funding 276 000 Euros, Before:co-PI of Germany branch. Now: Co-PI of Mexican branches.

2021–2023 JETSET-ERC Fellowship: Institut Für Theoretische Physik, Goethe Universität.

2019–2021 Black Hole Cam-EHT Fellowship: Institut Für Theoretische Physik, Goethe Universität.

2017–2019 CONACYT Postdoc abroad Fellowship: at Universitat de València.

2016–2017 ExaHyPE-Horizon 2020 EU Fellowship: Institut Für Theoretische Physik, Goethe Universität.

2015–2017 DGAPA postdoc Fellowship: at Institute of Astronomy, UNAM

2008–2010 Fellowship for Ph.D. studies: at the Institute of Physics and Mathematics, UMSNH ³, granted by the Mexican Council of Science CONACyT.

2010–2014 Fellowship for M.Sc studies: at the Institute of Physics and Mathematics, UMSNH, granted by the Mexican Council of Science CONACyT.

Computational Grants

6. **Co-PI** in the following projects of Laboratorio Nacional de Supercómputo del Sureste de México (the PIs are my students), **100,000 CPU-hours** in each project:
 - 202501016N - Evolución temporal de sistemas binarios de estrellas de neutrones con magnetohidrodinámica relativista, PI Omar Elías Velasco Castillo, Feb 2025– Feb 2026.
 - 202403062N - Polarización de la luz del jet relativista de M87 a escala sub-parsec, PI David Barrero González, Aug 2024 - Aug 2025.
 - 202304071C. -Evolución de discos magnetizados en objetos compactos, PI Juan José Zaldívar Vázquez, 2023-2024.
 - 202501006N - Estrellas mixtas relativistas con ecuaciones de estado realistas, PI Daniel Núñez Trigueros, Feb 2025 - Feb 2026.
5. **Co-PI** Estudio de la emisión electromagnética en multifrecuencias de Blazares: Comparación teórica-observacional, LAMOD-UNAM cluster. **1 Million CPU-hours**. 2023-2025.
4. **PI** Evolución de discos magnetizados alrededor de agujero negros, LAMOD-UNAM cluster. **0.5 Million CPU-hours**. 2023-2024.
3. **PI** of the project **Long-term simulations of magnetized disks and jets around supermassive black-hole binaries in General Relativity** HAWK cluster, High-Performance Computing Center Stuttgart, Stuttgart, Germany. **100 Millions CPU-hours**. 2023-2024.
2. **PI** of the project **Large-scale numerical-relativity simulations of tilted black hole-torus system**, project-ID: FI-2019-2-0043, **3 Millions CPU-hours**, Estimated cost: 2959 euros, Cluster: Barcelona Supercomputing center-MareNostrum 4.

SECIHTI: Secretaría de Ciencia, Humanidades, Tecnología e Innovación

<https://newfunfico.wixsite.com/newfunfico/universidad-mexico>

UMSNH: Universidad Michoacana de San Nicolás de Hidalgo

1. Microphysical aspects of binary neutron star mergers (BNSMIC), HAWK cluster, High-Performance Computing Center Stuttgart, Stuttgart, Germany. **100 Millions CPU-hours**. PI Luciano Rezzolla.

Publications

Publications.....

- [1] Navarrete, C., Vázquez-Chávez, F., **Cruz-Osorio**, A., & Ortiz, N. Testing Black Hole Spacetimes with the S2 Star Orbit: A Bayesian Comparison. *Mon. Not. R. Astron. Soc.*, **546(3)**:1–19, pages 1–19. Jan. 2026
- [2] Megale, R., **Cruz-Osorio**, A., Ficarra, G., et al. Measuring the properties of homogeneous turbulence in curved spacetimes. *Mon. Not. R. Astron. Soc.*, **544(2)**:2011–2023, pages 2011–2023. Oct. 2025
- [3] Imbrogno, M., Meringolo, C., **Cruz-Osorio**, A., et al. Turbulence and Magnetic Reconnection in Relativistic Multispecies Plasmas. *Astrophys. J. Lett.*, **990(2)**:L33, page L33. Sep. 2025
- [4] Santibañez-Armenta, M. J., Magallanes-Guijón, G., Mendoza, S., & **Cruz-Osorio**, A. Modelling the shadow of Sgr A* through an eclipsing black hole. *Mon. Not. R. Astron. Soc.: Letters*, **540(1)**:L54–L57, pages L54–L57. Jun. 2025
- [5] Acevedo-Muñoz, J. C., Lora-Clavijo, F. D., & **Cruz-Osorio**, A. Shadow and gravitational lensing produced by the nonlinear accretion of a scalar field onto a black hole. *Phys. Rev. D*, **111(8)**:084022, page 084022. Apr. 2025
- [6] Valdez-Alvarado, S., **Cruz-Osorio**, A., Dávila, J. M., Arturo Ureña-López, L., & Becerril, R. Observations of compact stars and fermion-boson stars with a quartic self-interaction. *Classical and Quantum Gravity*, **42(8)**:085001, page 085001. Apr. 2025
- [7] Moriyama, K., **Cruz-Osorio**, A., Mizuno, Y., Dihingia, I. K., & Uniyal, A. Black hole accretion and radiation variability in general relativistic magnetohydrodynamic simulations with Rezzolla–Zhidenko spacetime. *Astronomy and Astrophysics*, **694**:A135, page A135. Feb. 2025
- [8] Nathanail, A., Mizuno, Y., Contopoulos, I., et al. The impact of resistivity on the variability of black hole accretion flows. *Astronomy and Astrophysics*, **693**:A56, page A56. Jan. 2025
- [9] Imbrogno, M., Meringolo, C., Servidio, S., et al. Long-lived Equilibria in Kinetic Astrophysical Plasma Turbulence. *Astrophys. J. Lett.*, **972(1)**:L5, page L5. Sep. 2024
- [10] Gimeno-Soler, S., Pimentel, O. M., Lora-Clavijo, F. D., **Cruz-Osorio**, A., & Font, J. A. Magnetized tori with magnetic polarization around Kerr black holes: Variable angular momentum disks. *Phys. Rev. D*, **110(2)**:023023, page 023023. Jul. 2024
- [11] Zhang, M., Mizuno, Y., Fromm, C. M., Younsi, Z., & **Cruz-Osorio**, A. Impacts of nonthermal emission on the images of a black hole shadow and extended jets in two-temperature GRMHD simulations. *Astronomy and Astrophysics*, **687**:A88, page A88. Jul. 2024
- [12] Moriyama, K., **Cruz-Osorio**, A., Mizuno, Y., et al. Future Prospects for Constraining Black Hole Spacetime: Horizon-scale Variability of Astrophysical Jets. *Astrophys. J.*, **960(2)**:106, page 106. Jan. 2024
- [13] Kocherlakota, P., Narayan, R., Chatterjee, K., **Cruz-Osorio**, A., & Mizuno, Y. Toward General Relativistic Magnetohydrodynamics Simulations in Stationary Nonvacuum Spacetimes. *Astrophys. J. Lett.*, **956(1)**:L11, page L11. Oct. 2023
- [14] **Cruz-Osorio**, A., Rezzolla, L., Lora-Clavijo, F. D., et al. Bondi-Hoyle-Lyttleton accretion onto a rotating black hole with ultralight scalar hair. *Journal of Cosmology and Astroparticle Physics*, **2023(8)**:057, page 057. Aug. 2023

- [15] Mpisketzis, V., Duqué, R., Nathanail, A., **Cruz-Osorio**, A., & Rezzolla, L. Impact of anisotropic ejecta on jet dynamics and afterglow emission in binary neutron-star mergers. *Mon. Not. R. Astron. Soc.*, **527(3)**:9159–9175, pages 9159–9175. Jan. 2024
- [16] Röder, J., **Cruz-Osorio**, A., Fromm, C. M., et al. Probing the spacetime and accretion model for the Galactic Center: Comparison of Kerr and dilaton black hole shadows. *Astronomy and Astrophysics*, **671**:A143, page A143. Mar. 2023
- [17] Meringolo, C., **Cruz-Osorio**, A., Rezzolla, L., & Servidio, S. Microphysical Plasma Relations from Special-relativistic Turbulence. *Astrophys. J.*, **944(2)**:122, page 122. Feb. 2023
- [18] Fromm, C. M., **Cruz-Osorio**, A., Mizuno, Y., et al. Impact of non-thermal particles on the spectral and structural properties of M87. *Astronomy and Astrophysics*, **660**:A107, page A107. Apr. 2022
- [19] **Cruz-Osorio**, A., Fromm, C. M., Mizuno, Y., et al. State-of-the-art energetic and morphological modelling of the launching site of the M87 jet. *Nature Astronomy*, **6**:103–108, pages 103–108. Jan. 2022
- [20] **Cruz-Osorio**, A., Lora-Clavijo, F. D., & Herdeiro, C. GW190521 formation scenarios via relativistic accretion. *Journal of Cosmology and Astroparticle Physics*, **2021(07)**:032, page 032. Jul. 2021
- [21] **Cruz-Osorio**, A., Gimeno-Soler, S., Font, J. A., De Laurentis, M., & Mendoza, S. Magnetized discs and photon rings around yukawa-like black holes. *Phys. Rev. D*, **103(12)**:124009, page 124009. Jun. 2021
- [22] **Cruz-Osorio**, A., & Rezzolla, L. Common-envelope Dynamics of a Stellar-mass Black Hole: General Relativistic Simulations. *Astrophys. J.*, **894(2)**:147, page 147. May 2020
- [23] **Cruz-Osorio**, A., Gimeno-Soler, S., & Font, J. A. Non-linear evolutions of magnetized thick discs around black holes: dependence on the initial data. *Mon. Not. R. Astron. Soc.*, **492(4)**:5730–5742, pages 5730–5742. Mar. 2020
- [24] Arroyo-Chávez, G., **Cruz-Osorio**, A., Lora-Clavijo, F. D., Campuzano Vargas, C., & García Mora, L. A. Neutron and quark stars: constraining the parameters for simple EoS using the GW170817. *Astrophys Space Sci.*, **365(2)**:43, page 43. Feb. 2020
- [25] Becerra-Vergara, E. A., Mojica, S., Lora-Clavijo, F. D., & **Cruz-Osorio**, A. Anisotropic quark stars with an interacting quark equation of state. *Phys. Rev. D*, **100(10)**:103006, page 103006. Nov. 2019
- [26] **Cruz-Osorio**, A., Sánchez-Salcedo, F. J., & Lora-Clavijo, F. D. Relativistic Bondi-Hoyle-Lyttleton accretion in the presence of small rigid bodies around a black hole. *Mon. Not. R. Astron. Soc.*, **471(3)**:3127–3134, pages 3127–3134. Nov. 2017
- [27] **Cruz-Osorio**, A., & Lora-Clavijo, F. D. Non-axisymmetric relativistic wind accretion with velocity gradients on to a rotating black hole. *Mon. Not. R. Astron. Soc.*, **460(3)**:3193–3201, pages 3193–3201. Aug. 2016
- [28] González-Avilés, J. J., **Cruz-Osorio**, A., Lora-Clavijo, F. D., & Guzmán, F. S. Newtonian CAFE: a new ideal MHD code to study the solar atmosphere. *Mon. Not. R. Astron. Soc.*, **454(2)**:1871–1885, pages 1871–1885. Dec. 2015
- [29] Lora-Clavijo, F. D., **Cruz-Osorio**, A., & Moreno Méndez, E. Relativistic Bondi-Hoyle-Lyttleton Accretion onto a Rotating Black Hole: Density Gradients. *Astrophys. J., Supp.*, **219(2)**:30, page 30. Aug. 2015
- [30] Lora-Clavijo, F. D., **Cruz-Osorio**, A., & Guzmán, F. S. CAFE: A New Relativistic MHD Code. *Astrophys. J., Supp.*, **218(2)**:24, page 24. Jun. 2015
- [31] Lora-Clavijo, F. D., Guzmán, F. S., & **Cruz-Osorio**, A. PBH mass growth through radial accretion during the radiation dominated era. *Journal of Cosmology and Astroparticle Physics*, **2013(12)**:015, page 015. Dec. 2013

- [32] **Cruz-Osorio**, A., Lora-Clavijo, F. D., & Guzmán, F. S. Is the flip-flop behaviour of accretion shock cones on to black holes an effect of coordinates? *Mon. Not. R. Astron. Soc.*, **426**(1):732–738, pages 732–738. Oct. 2012
- [33] **Cruz-Osorio**, A., Guzmán, F. S., & Lora-Clavijo, F. D. Scalar field dark matter: behavior around black holes. *Journal of Cosmology and Astroparticle Physics*, **2011**(6):029, page 029. Jun. 2011
- [34] Lora-Clavijo, F. D., **Cruz-Osorio**, A., & Guzmán, F. S. Evolution of a massless test scalar field on boson star space-times. *Phys. Rev. D*, **82**(2):023005, page 023005. Jul. 2010
- [35] **Cruz-Osorio**, A., Gonzalez-Juarez, A., Guzman, F. S., & Lora-Clavijo, F. D. Numerical solution of the wave equation on particular space-times using CMC slices and scri-fixing conformal compactification. *Rev. Mex. Fis.*, **56**:456–468, pages 456–468. 2010
- [36] Bernshteyn, V., Conroy, N. S., Bauböck, M., et al. Ring Asymmetry and Spin in M87*. *The Astrophysical Journal*, **1000**(2):231, page 231. Apr. 2026
- [37] Saurabh, Müller, H., von Fellenberg, S. D., et al. Probing jet base emission of M87* with the 2021 Event Horizon Telescope observations. *Astronomy and Astrophysics*, **706**:A27, page A27. Jan. 2026
- [38] Gómez, J. L., Cho, I., Traianou, E., et al. Spatially resolved polarization swings in the supermassive binary black hole candidate OJ 287 with first Event Horizon Telescope observations. *Astronomy and Astrophysics*, **705**:A23, page A23. Jan. 2026
- [39] Event Horizon Telescope Collaboration, Akiyama, K., Albentosa-Ruíz, E., et al. Horizon-scale variability of M87* from 2017/2021 EHT observations. *Astronomy and Astrophysics*, **704**:A91, page A91. Dec. 2025
- [40] Dahale, R., Cho, I., Moriyama, K., et al. Origin of the ring ellipticity in the black hole images of M87*. *Astronomy and Astrophysics*, **699**:A279, page A279. Jul. 2025
- [41] Goddi, C., Carlos, D. F., Crew, G. B., et al. First polarization study of the M87 jet and active galactic nuclei at submillimeter wavelengths with ALMA. *Astronomy and Astrophysics*, **699**:A265, page A265. Jul. 2025
- [42] Röder, J., Wielgus, M., Lobanov, A. P., et al. A multifrequency study of sub-parsec jets with the Event Horizon Telescope. *Astronomy and Astrophysics*, **695**:A233, page A233. Mar. 2025
- [43] Event Horizon Telescope Collaboration, Akiyama, K., Albentosa-Ruíz, E., et al. The persistent shadow of the supermassive black hole of M87: II. Model comparisons and theoretical interpretations. *Astronomy and Astrophysics*, **693**:A265, page A265. Jan. 2025
- [44] Baczkó, A.-K., Kadler, M., Ros, E., et al. The putative center in NGC 1052. *Astronomy and Astrophysics*, **692**:A205, page A205. Dec. 2024
- [45] Algaba, J. C., Baloković, M., Chandra, S., et al. Broadband multi-wavelength properties of M87 during the 2018 EHT campaign including a very high energy flaring episode. *Astronomy and Astrophysics*, **692**:A140, page A140. Dec. 2024
- [46] Raymond, A. W., Doeleman, S. S., Asada, K., et al. First very long baseline interferometry detections at 870 m. *The Astronomical Journal*, **168**(3):130, page 130. aug 2024
- [47] Event Horizon Telescope Collaboration, Akiyama, K., Alberdi, A., et al. First Sagittarius A* Event Horizon Telescope Results. VIII. Physical Interpretation of the Polarized Ring. *Astrophys. J. Lett.*, **964**(2):L26, page L26. Apr. 2024
- [48] Event Horizon Telescope Collaboration, Akiyama, K., Alberdi, A., et al. First Sagittarius A* Event Horizon Telescope Results. VII. Polarization of the Ring. *Astrophys. J. Lett.*, **964**(2):L25, page L25. Apr. 2024

- [49] Parascos, G. F., Kim, J. Y., Wielgus, M., et al. Ordered magnetic fields around the 3C 84 central black hole. *Astronomy and Astrophysics*, 682:L3, page L3. Feb. 2024
- [50] Event Horizon Telescope Collaboration, Akiyama, K., Alberdi, A., et al. The persistent shadow of the supermassive black hole of M 87. I. Observations, calibration, imaging, and analysis. *Astronomy and Astrophysics*, 681:A79, page A79. Jan. 2024
- [51] Roelofs, F., Johnson, M. D., Chael, A., et al. Polarimetric Geometric Modeling for mm-VLBI Observations of Black Holes. *Astrophys. J. Lett.*, 957(2):L21, page L21. Nov. 2023
- [52] Event Horizon Telescope Collaboration, Akiyama, K., Alberdi, A., et al. First M87 Event Horizon Telescope Results. IX. Detection of Near-horizon Circular Polarization. *Astrophys. J. Lett.*, 957(2):L20, page L20. Nov. 2023
- [53] Torne, P., Liu, K., Eatough, R. P., et al. A search for pulsars around sgr a* in the first event horizon telescope data set. *The Astrophysical Journal*, 959(1):14, page 14. nov 2023
- [54] Prather, B. S., Dexter, J., Moscibrodzka, M., et al. Comparison of Polarized Radiative Transfer Codes Used by the EHT Collaboration. *Astrophys. J.*, 950(1):35, page 35. Jun. 2023
- [55] Jorstad, S., Wielgus, M., Lico, R., et al. The Event Horizon Telescope Image of the Quasar NRAO 530. *Astrophys. J.*, 943(2):170, page 170. Feb. 2023
- [56] Broderick, A., Pesce, D., Gold, R., et al. The Photon Ring in M87*. *Astrophys. J.*, 935(1):61, page 61. Aug. 2022
- [57] Issaoun, S., Wielgus, M., & Event Horizon Telescope Collaboration. Resolving the inner parsec of the blazar J1924–2914 with the Event Horizon Telescope. *Astrophys. J.*, 934(2):21, page 21. Jul. 2022
- [58] Broderick, A., & Event Horizon Telescope Collaboration. Characterizing and Mitigating Intraday Variability: Reconstructing Source Structure in Accreting Black Holes with mm-VLBI. *Astrophys. J. Lett.*, 930(2):L21, page L21. May 2022
- [59] Georgiev, B., & Event Horizon Telescope Collaboration. A Universal Power-law Prescription for Variability from Synthetic Images of Black Hole Accretion Flows. *Astrophys. J. Lett.*, 930(2):L20, page L20. May 2022
- [60] Wielgus, M., & Event Horizon Telescope Collaboration. Millimeter Light Curves of Sagittarius A* Observed during the 2017 Event Horizon Telescope Campaign. *Astrophys. J. Lett.*, 930(2):L19, page L19. May 2022
- [61] Farah, J., & Event Horizon Telescope Collaboration. Selective Dynamical Imaging of Interferometric Data. *Astrophys. J. Lett.*, 930(2):L18, page L18. May 2022
- [62] Event Horizon Telescope Collaboration, Akiyama, K., Algaba, J. C., Alberdi, A., & Alef, W. First Sagittarius A* Event Horizon Telescope Results. VI. Testing the Black Hole Metric. *Astrophys. J. Lett.*, 930(2):L17, page L17. May 2022
- [63] Event Horizon Telescope Collaboration, Akiyama, K., Algaba, J. C., Alberdi, A., & Alef, W. First Sagittarius A* Event Horizon Telescope Results. V. Testing Astrophysical Models of the Galactic Center Black Hole. *Astrophys. J. Lett.*, 930(2):L16, page L16. May 2022
- [64] Event Horizon Telescope Collaboration, Akiyama, K., Algaba, J. C., Alberdi, A., & Alef, W. First Sagittarius A* Event Horizon Telescope Results. IV. Variability, Morphology, and Black Hole Mass. *Astrophys. J. Lett.*, 930(2):L15, page L15. May 2022
- [65] Event Horizon Telescope Collaboration, Akiyama, K., Algaba, J. C., Alberdi, A., & Alef, W. First Sagittarius A* Event Horizon Telescope Results. III. Imaging of the Galactic Center Supermassive Black Hole. *Astrophys. J. Lett.*, 930(2):L14, page L14. May 2022

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- [67] Event Horizon Telescope Collaboration, Akiyama, K., Algaba, J. C., Alberdi, A., & Alef, W. First Sagittarius A* Event Horizon Telescope Results. I. The Shadow of the Supermassive Black Hole in the Center of the Milky Way. *Astrophys. J. Lett.*, **930(2)**:L12, page L12. May 2022
- [68] Satapathy, K., Psaltis, D., Özel, F., et al. The Variability of the Black Hole Image in M87 at the Dynamical Timescale. *Astrophys. J.*, **925(1)**:13, page 13. Jan. 2022
- [69] Janssen, M., Falcke, H., Kadler, M., et al. Event Horizon Telescope observations of the jet launching and collimation in Centaurus A. *Nature Astronomy*, **5**:1017–1028, pages 1017–1028. Jul. 2021
- [70] Kocherlakota, P., Rezzolla, L., Falcke, H., et al. Constraints on black-hole charges with the 2017 EHT observations of M87*. *Phys. Rev. D*, **103(10)**:104047, page 104047. May 2021
- [71] Narayan, R., Palumbo, D. C. M., Johnson, M. D., et al. The Polarized Image of a Synchrotron-emitting Ring of Gas Orbiting a Black Hole. *Astrophys. J.*, **912(1)**:35, page 35. May 2021
- [72] EHT MWL Science Working Group, Algaba, J. C., Anczarski, J., et al. Broadband Multi-wavelength Properties of M87 during the 2017 Event Horizon Telescope Campaign. *Astrophys. J. Lett.*, **911(1)**:L11, page L11. Apr. 2021
- [73] Goddi, C., Martí-Vidal, I., Messias, H., et al. Polarimetric Properties of Event Horizon Telescope Targets from ALMA. *Astrophys. J. Lett.*, **910(1)**:L14, page L14. Mar. 2021
- [74] Event Horizon Telescope Collaboration, Akiyama, K., Algaba, J. C., et al. First M87 Event Horizon Telescope Results. VIII. Magnetic Field Structure near The Event Horizon. *Astrophys. J. Lett.*, **910(1)**:L13, page L13. Mar. 2021
- [75] Event Horizon Telescope Collaboration, Akiyama, K., Algaba, J. C., et al. First M87 Event Horizon Telescope Results. VII. Polarization of the Ring. *Astrophys. J. Lett.*, **910(1)**:L12, page L12. Mar. 2021
- [76] Sanchez-Bermudez, J., **Cruz-Osorio**, A., Barrera-Ballesteros, J. K., Alberdi, A., & Schödel, R., Aug. 2024, in *Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series*, Vol. 13095, *Optical and Infrared Interferometry and Imaging IX*, ed. J. Kammerer, S. Sallum, & J. Sanchez-Bermudez, page 1309519
- [77] Chabanov, M., **Cruz-Osorio**, A., Ecker, C., et al., Apr. 2024, in *High Performance Computing in Science and Engineering '22*, Springer Nature Switzerland, ed. W. E. Nagel, D. H. Kröner, & M. M. Resch, page 19
- [78] Röder, J., **Cruz-Osorio**, A., Fromm, C. M., et al., Jan. 2022, in *European VLBI Network Mini-Symposium and Users' Meeting 2021*, page 24
- [79] **Cruz-Osorio**, A., Lora-Clavijo, F. D., & Guzmán, F. S., Jul. 2013, in *American Institute of Physics Conference Series*, Vol. 1548, *IX Mexican School on Gravitation and Mathematical Physics: Cosmology for the XXIst Century: Gravitation and Mathematical Physics Division of the Mexican Physical Society DGFM-SMF*, ed. L. A. Uren-López, R. Becerril-Bárcenas, & R. Linares-Romero, pages 323–327
- [80] **Cruz-Osorio**, A., Lora-Clavijo, F. D., & Guzmán, F. S., Jul. 2010, in *American Institute of Physics Conference Series*, Vol. 1256, *Gravitational Physics: Testing Gravity from Submillimeter to Cosmic*, ed. H. A. Morales-Tecotl, L. A. Urena-Lopez, R. Linares-Romero, & H. H. Garcia-Compean, pages 311–317

Teaching

2026-1 Relatividad General Avanzada y Aplicaciones Astrofísicas, MSc. in Astrophysics, IA-UNAM, México.

2026-1 SI⁴ Procesos de acreción en agujeros negros, MSc. in Astrophysics, IA-UNAM, México.
 2026-1 Relativity, Faculty of Science(Physic), UNAM, México.
 2026-1 TSA II⁵: Black Hole Shdows, Faculty of Science(Physic), UNAM, México.
 2025-2 SG⁶ Modeling the Sub-Parsec Jet of Centaurus A, MSc. in Astrophysics, IA-UNAM, México.
 2025-1 Problemas contemporáneos de Astrofísica Computacional, MSc. in Astrophysics, IA-UNAM, México.
 2025-1 SI Introducción a la astrofísica relativista, SI, MSc. in Astrophysics , IA-UNAM, México.
 2024-2 SI Modelado de la emision de AGNs con jet, SI, MSc. in Astrophysics, IA-UNAM, México.
 2024-2 Numerical Relativistic Magnetohydrodynamics, MSc. in Astrophysics, IA-UNAM, México.
 2024-2 Relativity, Faculty of Science (Physic), UNAM, México.
 2024-1 Quantum Mechanics, M. Sc. in Astrophysics, IA-UNAM, México.
 2022-2 Hydrodynamics and Magnetohydrodynamics, M. Sc. in Physics, ITP ⁷, Germany. [Notes here](#)
 2016-2 Relativity, Faculty of Science (Physic), UNAM, México.
 2016-2 Electrodynamics, M. Sc. in Astrophysics program, Institute of Astronomy-UNAM, México.
 2015-2 Hydrodynamics in general relativity, M. Sc. in Astrophysics, IA-UNAM, México.

Students Mentoring

Donaldo E. Mora (M.Sc. in Astrophysics)/"Mención Honorífica"	22.02.2026
<i>Instituto de Astronomía, UNAM, México</i>	<i>co-supervisor: Erika Benítez Lizaola</i>
A GRRT model for the relativistic jet morphology and low energy SED of M87	
Giuseppe Riveccio (M.Sc. in Physics)	17.11.2023
<i>Università degli Studi di Napoli Federico II, Italy</i>	<i>co-supervisor: Mariafelicia De Laurentis</i>
Tracking hotspots around Sgr A*	
Mauricio Ortiz (M.Sc. in Astrophysics) with Summa cum laude	04.09.2023
<i>Institut für Theoretische Physik, University of Cologne, Germany</i>	<i>co-supervisor: Claus Kiefer</i>
General Relativistic Magnetohydrodynamic Simulations of Relativistic Jets Interacting with Stellar Winds	
Jan Röder (M.Sc. in Astrophysics) with Summa cum laude	14.04.2020
<i>Institut für Theoretische Physik, Goethe Universität Frankfurt, Germany</i>	<i>co-supervisor: Luciano Rezzolla</i>
Comparison of Kerr and dilaton black hole shadows: Impact of non-thermal emission	
Luis García (B.Sc. in Physics)[24]	12.03.2019
<i>Faculty of Physics, UNAM, México</i>	
Quark star using MIT bag model	
Griselda Arroyo Chávez (B.Sc. in Physic)/"Mención Honorífica" [24]	26.01.2018
<i>Faculty of Physics, Universidad Veracruzana, México</i>	<i>co-supervisor: C. Campuzano</i>
Observational and numerical comparability of neutron stars models using a polytropic EoS	
Summer research visiting.....	
○ María Fernanda Osorno Tinoco (Mención Honorífica), Karol Guadalupe Osorno Tinoco y Gabriel Luna Morett. Estancia Corta 2024 del Programa Jóvenes hacia la Investigación, DGDC-UNAM, 03.jun.2024 – 02.aug.2024.	
○ Kevin Eduardo Reyes Valle (Universidad Autónoma de Sinaloa), Estancia de Verano de Investigación, IA-UNAM, 01.aug.2025–29.aug.2025	

Examiner

Thesis Defence:

- MSc., Maximiliano del Carpio Cruz, Soluciones Analíticas al problema de superficies de trabajo internas en jets no-relativistas, 06.Nov.2025, IA-UNAM, México.
- BSc., Edgar Pastrana Calderón, Refinamiento fijo de malla numérica en hidrodinámica relativista, 29.09.2025, Facultad de Ciencias UNAM.

SI: Seminario de Investigación

TSA II: Temas Selectos de Astrofísica II

SG: Seminario de Graduación

ITP: Institute of Theoretical Physics, Goethe-Universität Frankfurt

- *PhD*, Jessica Luna Cervantes, Multiwavelength characterization of jetted AGNs through the SED modeling, 08.08.2025, IA-UNAM, México.
- *BSc.*, Jesús Aaron Flores Prudente, Estrellas de Dirac: Estabilidad dinámica de los estados excitados, 10.06.2025.
- *BSc.*, Ramón Tamayo Andrade, Análisis del movimiento geodésico en el límite extremal de un agujero negro en la teoría Einstein-Maxwell-Dilatón, 22.01.2025.
- *PhD*, Milton Jair Santibañez Armenta, Deflection of massive and massless particles and some applications to black hole shadows, 13.01.2025, IA-UNAM, México.
- *PhD*, Fernando Josue Ureña Mena, Multifrequency study of Very High Energy emitter Active Galactic Nuclei observed with HAWC, 03.Dec.2024, INAOE, México.
- *MSc.*, Omar Vilchis Alcántar, Unveiling the Magneto-ionic structures of the radio relic of MACS07+3745, 28.Nov.2024, IRyA-UNAM, México.
- *BSc.*, Enrique Galicia Pineda, Modelado de estrellas compactas con el programa Aztekas, 15.April.2024, Facultad de Ciencias UNAM, México.
- *BSc.*, Fernando Vázquez Chávez, Orbitas estelares como herramienta para estudiar la naturaleza del espacio-tiempo, 04.March.2024, Facultad de Ciencias UNAM, México.
- *BSc.*, German David Prada Méndez, Análisis del espectro de emisión proveniente de un toro magnetizado alrededor de singularidades desnudas con deformación cuadrupolar, 17.February.2023, Faculty of Physics, Universidad Industrial de Santander, Colombia.
- *BSc.*, Jennyfer Camila Acevedo Muñoz, Análisis numérico de la interacción del sistema agujero negro–campo escalar en un espacio-tiempo dinámico (17.February.2023), Faculty of Physics, Universidad Industrial de Santander, Colombia.
- *MSc.*, Juan Manuel Velásquez Cadavid, Simulation of the intensity map in magnetically polarized accretion disks around Kerr black holes (13.October.2022), Master of Applied Mathematics, Faculty of Physics, Universidad Industrial de Santander, Colombia.
- *BSc.*, Cédric Jockel, Modelling of plasma accretion onto black hole mimickers (20.October.2021), Institut für Theoretische Physik, Johann Wolfgang Goethe -Universität Frankfurt am Main, Germany.
- *BSc.*, Jose Miguel Amado Dugarte, Numerical study of relativistic magnetohydrodynamic laminar currents in a magnetically polarized fluid (21.July.2021), Faculty of Physics, Universidad Industrial de Santander, Colombia.
- *MSc.*, Jan Röder, Comparison of Kerr and dilaton black hole shadows: Impact of non-thermal emission (13.April.2021), Institut für Theoretische Physik, Johann Wolfgang Goethe -Universität Frankfurt am Main, Germany.
- *MSc.*, Manuel Eduardo de la Cruz Hernández, Modelling of work surfaces in 1D astrophysical jets at the relativistic strong shock limit (31.Jan.2019), Institute of Astronomy, UNAM, México.
- *MSc.*, Alejandro Aguayo Ortiz, A direct Primitive Variable Recovery Scheme for hyperbolic conservative equations (27.Jul.2018), Institute of Astronomy, UNAM, México.

Thesis Project:

- *PhD*, Fernando Rojas Contreras, May 29, 2025, Estudio Girocinético del Transporte Anómalo Producto de Ondas de Deriva en Propulsores Hall, PhD on Physics, ICN-UNAM, México.
- *BSc.*, German David Prada Méndez, Análisis del espectro de emisión proveniente de un toro magnetizado alrededor de singularidades desnudas con deformación cuadrupolar (22.August.2022), Faculty of Physics, Universidad Industrial de Santander, Colombia.
- *PhD*, Gustavo Magallanes Guijón, December 3, 2021, Statistical and computational methods in inference binary black holes, PhD on Astrophysics, IA-UNAM, México.
- *MSc.*, Juan Manuel Velásquez Cadavid, Simulation of the intensity map in magnetically polarized accretion disks around Kerr black holes (04.December.2020), Master of Applied Mathematics, Faculty of Physics, Universidad Industrial de Santander, Colombia.

Tutor committee:

- *MSc.* Zenaida Berenice Sánchez Méndez (2025-2 – now), Astrophysics, IA-UNAM, México.
- *MSc.* Jeoshua Alejandro Licea Garcia (2025-1 – now), Physics, ICN-UNAM, México.

- MSc. Axel Morales Buendía (2025-1 – now), Physics, ICN-UNAM, México.
- MSc. Yael Morales Venegas (2025-1 – now), Astrophysics, IA-UNAM, México.
- MSc. César Orlando Navarrete Zavala (2024-2 – 2026-1), Astrophysics, ICN-UNAM, México.
- MSc. Anthony Brandon Arenas Martínez (2024-2–now), Astrophysics, IA-UNAM, México.
- PhD. Alejandro Aguayo Ortiz (2018 –2022), Astrophysics, IA-UNAM⁸, México.

Talks

Invited talks and invited Lectures.....

Microscopic analysis of astrophysical plasmas: Special Relativistic Particle In Cell Simulation

Invited Talk, Taller de Fluidos Relativistas,
UAM, Cuajimalpa, CDMX, México. Nov. 14th 2025

Computación de Alto Rendimiento en Astrofísica Relativista

Invited Talk, Congreso Internacional de Supercómputo 2025,
UNAM, CDMX, México. Sep. 2nd 2025

Magnetized Accretion Disks and Jet Launching: Nonlinear Evolution and Insights from EHT Observations

Invited Talk, 25th Guillermo Haro Workshop,
INAOE, Tonantzintla, Puebla. Jul. 28th 2025

Modelado de sombras de agujeros negros supermasivos

Invited Lecture, XV Taller de la División de Gravitación y Física-Matemática-SMF,
León, Guanajuato, México. Oct. 21-24 2024

Cocinando sombras cósmicas usando simulaciones de magnetohidrodinámica relativista

Invited Talk, Escuela de Verano del INAOE 2024,
Tonantzintla, Puebla, México. Jul 5th 2024

Agujeros Negros

Invited Lecture, Primera Escuela de Verano, Observatorio Astronómico Nacional Tonantzintla 2024,
Tonantzintla, Puebla, México. Jun. 30-Jul. 5 2024

Using ngVLA to probe gravity, plasma properties and particle acceleration

Invited Talk, First Mexican meeting in route to the Next Generation VLA, Morelia, México. Sep. 25-27 2023

Interpretando la imagen de la sombra de Sagitario A* utilizando relatividad general

Plenary Talk, LXV Congreso Nacional de Física, Zacatecas, México. Oct 4 2022

Shadow of the supermassive black hole in M87

Review Talk, Growing Black Holes: Accretion & Mergers In memory of Sergio Colafrancesco
Kathmandu, Nepal, May 15-20, 2022

GW190521 formation scenarios via relativistic accretion

Engineering Week, Faculty of Higher Studies "Aragón" -UNAM, Mexico City, México. October 28th 2021.

Seminars and Colloquium.....

Sombras de Agujeros Negros con Python

Seminario de Física Gravitacional y Geométrica de la Facultad de Física,
Universidad Veracruzana, Xalapa, México, May. 28th 2025

Shadow from nonlinear accretion of a scalar field onto a black hole

AstroLunch de Astrofísica Relativista, CDMX, México, Jan. 20th 2025

Shadow from nonlinear accretion of a scalar field onto a black hole

AstroCoffee Seminar, ITP — Goethe University, Frankfurt am Main, Germany, Dec 2nd 2024.

Modeling the hot spots of the black hole shadow of SgrA*

Seminar of Dipartimento di Fisica, Università della Calabria, Cosenza, Italy, Nov 27th 2024.

Modeling the supermassive black hole shadow SgrA*

Seminar of Dipartimento di Fisica, Università Federico II, Napoli, Italy, Nov 25th 2024.

IA-UNAM: Instituto de Astronomía, Universidad Nacional Autónoma de México

Sombras de agujeros negros: Un nuevo horizonte para las teorías de la gravedad

Seminario de Astrofísica del INAOE, Tonantzintla, Puebla, México. Sep 20th 2024.

Testing modified theory of gravity with shadows

Seminario del Departamento de Gravitación y Teoría de Campos del ICN, UNAM

CDMX, México. Sep 12 2024.

Modeling multi-wavelength emission of relativistic jets with simulations

Seminar of Dipartimento di Fisica, Università della Calabria, Cosenza, Italy, Jul 11th 2024.

Relativistic Jets as laboratory to test gravity, particle acceleration and plasma properties

Seminario "Jesús Reyes Corona", Instituto de Física, Benemérita Universidad Autónoma de Puebla, Puebla, México. Feb 23 2024.

Modelando la sombra de un agujero negro supermasivo: M87* y SgrA*

Seminario del departamento de Física, Facultad de Ciencias, UNAM, México. Feb 9th 2024.

An update on the supermassive black hole shadow M87*

Coloquio del Instituto de Astronomía, UNAM, México. Jan 24th 2024.

Jets relativistas bajo el microscopio: Efectos de la microfísica en la emisión electromagnética

Seminar group DAEC, Institute of Astronomy-UNAM, Mexico City, México. Nov 23th 2023.

Acceleration of non-thermal particles from turbulent plasmas

Coloquio del Instituto de Ciencias Físicas, UNAM, Cuernavaca, Morelos, Mexico. Jan 27 2023.

Interpretando la imagen de la sombra de Sagitario A* utilizando relatividad general
Seminario del Dep. de Física, Universidad Autónoma Metropolitana Iztapalapa, México. Oct 7th 2022

Interpretación teórica de la sombra del agujero negro supermasivo Sagitario A*
Coloquio del Instituto de Astronomía, UNAM, Mexico. Oct 6, 2022.

SgrA*: Our own black hole at the heart of the Milky Way
Seminario del Instituto de Física y Matemáticas/Facultad de Ciencias Físico-Matemáticas de la UMSNH

Morelia, Michoacán, México. May 27 2022

Jets relativistas:

Mecanismo de generación, aceleración de partículas y emisión electromagnética en multi-frecuencias

Coloquio del Instituto de Astronomía, UNAM, Mexico. May 11, 2022.

Electromagnetic emission from relativistic jets:

Modeling the launching and high energy emission using GRMHD simulations

Seminar, Instituto Avanzado de Cosmología, Mexico. April 1, 2022.

Multiwavelength emission of M87 jet from GRMHD simulations

Seminario "Dr. Jesús Reyes Corona", Instituto de Física—UAP, Mexico. February 4th 2022.

Understanding multi-wavelength emission of M87 jet using general relativistic MHD simulations

AstroLunch, The Netherlands Institute for Radio Astronomy, Netherlands. January 26th 2022.

Modeling Radiation Emission of M87 Jet with GRMHD Simulations

Dep. of Gravitation and Field theory seminar, Institute of Nuclear Physics-UNAM, México. Sep. 30th 2021.

General relativistic simulations of the common-envelope evolution

Physics Department Seminar, Universidad de la Sapienza Roma, Roma, Italy. March 17th, 2021.

Electromagnetic emissions from relativistic jets and magnetized disks

Multidisciplinary seminar, Facultad de ciencias,

Universidad Autónoma del Estado de México, Toluca, México. November 26th 2020

Relativistic Fluid Accretion Around a Black Hole

AstroCoffee seminar, Institute for Theoretical Physics, Goethe University,

Frankfurt, Germany. May 9th 2017

Fix Mesh Refinement in Cartesian coordinates

Institute of Astronomy-UNAM, Mexico City, México. June 14th 2016.

Newton (Siglo XVII)—Diplomado en Astrofísica

Institute of Astronomy-UNAM, Mexico City, México. May 12th 2016.

Relativistic Bondi-Hoyle accretion onto rotating black holes using CAFE code

Seminar of National Institute of Astrophysics, Optics and Electronics(INAOE)

Tonantzintla, Puebla, México. March 4th 2016.

Primordial black holes as seeds of supermassive black holes

Seminar of the Faculty of Physics, Universidad Veracruzana, Veracruz, México. February 4th 2016.

Relativistic accretion of supersonic winds around rotating black holes

Seminar group DATA, Institute of Astronomy-UNAM, Mexico City, México. January 19th 2016.

Contributed talks in conferences.....

Constraining Theories of Gravity Using Black Hole Shadows and Relativistic Jets

XXXIII Reunión Anual de la División de Gravitación y Física Matemática

CDMX, México, April. 24th 2025

High Performance Computing in Relativistic Astrophysics

1er BootCamp: Introducción a la Programación Científica IA-UNAM

CDMX, México, Jul. 30th 2024

Influence of Initial Accretion Disc properties on GRMHD Simulation Outcomes

The Event Horizon Telescope Collaboration meeting 2024

CDMX, México, May. 29th - 24th 2024

Relativistic jets as laboratory to test gravity, particle acceleration, and plasma properties.

Workshop on Fundamental Fields and Compact Objects: New Opportunities, ICF-UNAM, Morelos, México, Oct 4th - 6th 2023

Multi-wavelength emission of M87 jet using self-consistent electron-to-proton temperature

Workshop on Kinetic Models of Relativistic Plasmas, Trinity College, Dublin, Ireland, Feb. 27 – March 2, 2023.

Non-thermal emission at event horizon and jet launching scales

New Generation Event Horizon Telescope Meeting 2022. June 22-25 2022.

Modeling the limb brightening of M87 jet: Effects of electron temperature and magnetic energy

Event Horizon Telescope Meeting 2022. June 19-22 2022.

Modeling MWL emission with GRMHD and GRRT simulations

Event Horizon Telescope Meeting 2021. December 6-10 2021.

Non-thermal emission in SANE and MAD GRMHD simulations

Event Horizon Telescope SgrA Theory Symposium 2020. September 4th 2020.*

Non-linear evolution of magnetized-torus-BH: Comparison between magnetic field approaches

689. WE-Heraeus-Seminar on Accretion in strong gravity, Bad Honnef, Germany. February 4-8 2019.

Evolution of a Magnetized Thick disk around a highly rotating black hole

The European Einstein Toolkit meeting 2018, Lisbon, Portugal. September 13th 2018.

Consistent construction of a magnetised accretion disc around a highly rotating black hole

Spanish-Portuguese Relativity Meeting 2018, Palencia, Spain. September 4th 2018.

Special and General relativistic magnetohydrodynamics in ExaHyPE

ExaHyPE council Meeting, Garching, Germany. April 3th 2017.

CAFE a new relativistic MHD code

28th Texas Symposium on Relativistic Astrophysics, Geneva, Switzerland. December 2015.

Relativistic Bondi-Hoyle Accretion onto a Rotating Black Hole: Density Gradients

X Mexican School on Gravitation and Mathematical Physics, Playa del Carmen, México. December 2014.

Morelia, a new special relativistic MHD code

X workshop of the DGFM, Hidalgo, México. December 2013.

Morelia, a new special relativistic MHD code

XXV National Congress of Astronomy, Mexico city, México. November 2013.

Flip-flop instability of the shock cone in Bondi-Hoyle accretion

IX Mexican School on Gravitation and Mathematical Physics, Guadalajara, México. December 2012.

Flip-flop instability in the wind accretion around a rotating black hole

LV National Congress of Physics, Morelia, México. October 2012.

Bondi-Hoyle accretion onto Schwarzschild and Kerr black holes

XXIV National Congress of Astronomy, Guadalajara, México. September 2011.

Solution of a scalar field on a background of Schwarzschild containing the null infinity future

LIII Congress of Physics, Veracruz, México. October 2010.

Numerical solution of the wave equation on space-times with scri-fixing conformal compactifications

VIII Mexican School on Gravitation and Mathematical Physics of the DGFM-SMF,

Playa del Carmen, México. December 10th 2009

Poster Sessions.....

A model for M87 SED using a General Relativistic Radiative Transfer code

XXXIII Reunión Anual de la División de Gravitación y Física Matemática

CDMX, México, April. 24th 2025

A model for M87 SED using a General Relativistic Radiative Transfer code

XV Taller de la División de Gravitación y Física-Matemática-SMF,

León, Guanajuato, México. Oct. 21-24 2024. Authors: Donaldo Mora, Erika Benitez, Alejandro Cruz Osorio

Gravitational Waves From Long-Time Evolution Of Tilted Thick Disc Around A Rotating Black Hole

22nd International Conference on General Relativity and

Gravitation-13th Edoardo Amaldi Conference on Gravitational Waves, València, Spain. July 7th-12th 2019.

Stellar winds effect in Bondi-Hoyle accretion around Schwarzschild black hole

3rd Karl Schwarzschild Meeting - Gravity and the Gauge/Gravity Correspondence,

Frankfurt, Germany. July 24th-28th 2017.

Wind accretion on a Schwarzschild black hole in slab symmetry

LIV National Congress of Physics 2011, Yucatán, México. October 2011

Basic tests of general relativity hydrodynamics codes

VII School of the Gravitation and Mathematical Physics Division of the Mexican Physical Society,

Playa del Carmen, México. December 2006.

Lectures at Workshops.....

taller “Sombras de Agujeros Negros con Python”

Facultad de Física, Universidad Veracruzana,

Xalapa Ver., a 28 de mayo de 2025

Numerical methods for the study of relativistic astrophysics

Workshop on Numerical Relativity, Faculty of Physics, University of Veracruz,
Xalapa, Veracruz. November 18th-20th 2015.

Introduction to programming in Fortran 90

18 anniversary of the Instituto Tecnológico Superior de los Ríos,
Tabasco, México. November 26th & 27th 2014.

The science of science fiction

First Multidisciplinary Academic Week of the Instituto Tecnológico Superior de los Ríos,
Tabasco, México. November 17th & 18th 2014.

Las Leyes de Newton

Diplomado en Astrofísica, Instituto de Astronomía-UNAM,
Ciudad Universitaria, CDMX, a 14 de Junio del 2016

Outreach.....

Agujeros Negros Supermasivos: Observando lo Invisible

Voces de la Astronomía OAN-Tonantzintla, Tonantzintla, Puebla, México. Mar. 12 2026.

Coordinador de la carpa Agujeros negros: Entre bailes y sombras cósmicas

Noche de las Estrellas, Instituto de Astronomía-UNAM, Mexico city, México. Nov 29th 2025.

Ondas Gravitacionales y su impacto en la astrofísica moderna

El Universo los Viernes, Instituto de Astronomía-UNAM, Mexico city, México. Jun 20th 2025.

Talk

Platicando con un Astrónomo

Noche de las Estrellas, Instituto de Astronomía-UNAM, Mexico city, México. Nov 9th 2024.

Talk

Coordinador de la carpa: Astrofísica Relativista

Noche de las Estrellas, Instituto de Astronomía-UNAM, Mexico city, México. Nov 9th 2024.

Ondas Gravitacionales

Octubre Mes de la Astronomía, Sociedad Astronómica Queretana, Queretaro, México. Oct. 17 2024.

Talk

Observando sombras cósmicas con el EHT, un telescopio del tamaño de la tierra

Los Caminos de la Astronomía OAN-Tonantzintla, Tonantzintla, Puebla, México. Oct. 10 2024.

Talk

Sombras Cósmicas: El agujero negro en el centro de nuestra galaxia

Noche de Museos, Palacio de la Autonomía-UNAM, Mexico city, México. April 24th 2024.

Talk

Coordinador de la carpa: Astrofísica Relativista

Noche de las Estrellas, Instituto de Astronomía-UNAM, Mexico city, México. November 25th 2023

¡Agujeros Negros fantasticos y Como Encontrarlos!

Noche de las Estrellas, Instituto de Astronomía-UNAM, Mexico city, México. November 25th 2023.

Talk

La primera imagen de la sombra del agujero negro en el centro de nuestra galaxia

Conversatorio en el Museo de Arte e Historia de Guanajuato,
Leon, Guanajuato, México. Oct 14th 2023.

Talk

Agujeros negros y como encontrarlos sin morir en el intento

Seminario de Estudiantes del Posgrado en Astrofísica.,
UNAM, Ciudad de México, México. February 22th 2022.

Talk

Siluetas en la oscuridad: El agujero negro en el corazón de la Vía Láctea

Seminario del Club de Ciencias "Johann Carl Friedrich Gauss",
Facultad de Ingeniería Universidad Autónoma de Chihuahua. Chihuahua, México. November 10th 2022.

Talk

Telescopio del Horizonte de Eventos - El centro de la Vía Láctea

CONACYT, Mexico. August 2022.

Interview

Physik an der Goethe-Universität: Auszeichnung für herausragende Forschung

(Physics at the Goethe University: Award for outstanding research)

Goethe Universität, Frankfurt am Main, Germany. July 08th 2022.

Post

Congratulations to Dr. Cruz-Orsorio, laureate of the 2022 Frankfurt Physics Prize!

Goethe Universität, Frankfurt am Main, Germany. July 08th 2022.

Post

Astronomers reveal first image of the black hole at the heart of our galaxy

Event Horizon Telescope-Mexico, May 12th 2022.

Press Conference

Panelist of EHT Conference(Mexico) – YouTube

Simulando los chorros del agujero negro de M87, Minute 1:55:00

Coffee Break: Señal y Ruido, November 12th 2021, Spain.

Podcast

Coffee Break (YouTube) Ep341: Gravitondas; Monopolos; Fluidos en Roca Porosa; Agua en Galaxias; Agujeros Negros, Minute 2:08:35

Wie Schwarze Löcher ihren Jet zünden

UniReport Nr.6, Goethe Universität, Frankfurt am Main, Germany. December 9th 2021.

Press

Jet from giant galaxy M87: Computer modelling explains black hole observations

Goethe Universität, Frankfurt am Main, Germany. November 04th 2021.

Press Release

Nature Astronomy Press Release has 119 replicas in 36 countries.

Der Astrophysiker Alejandro Cruz Osorio im Porträt

UniReport Nr.5, Goethe Universität, Frankfurt am Main, Germany. October 12th 2017.

Press

The first image of a supermassive hole

"Jornadas de Ciencia y Matemáticas", Center for research and teaching of mathematics,

Morelia, México. October 13th 2020.

Talk

A supermassive black hole as the engine of our galaxy

Astronomical Society of Mexico, Mexico city, México. March 30th 2016.

Talk

A supermassive black hole at the center of our galaxy!

Night of Stars, Institute of Astronomy-UNAM, Mexico city, México. November 28th 2015.

Talk

Talking with Astronomers

Night of Stars, Institute of Astronomy-UNAM, Mexico city, México. November 28th 2015.

Talk

What is a black hole?

Feast of Sciences and Humanities, UNAM UNIVERSUM-Museum,

Mexico city, México. October 17th 2015.

Talk

LIGHT: our eyes throughout the universe

22 National Week of Science and Technology, International Year of Light,

Instituto Tecnológico Superior Huichapan, Hidalgo, México. October 12th 2015.

Talk

Nota sobre magnetohidrodinámica

Radio UNAM 96.1 FM, Programa: Primer Movimiento min:1:54,

Mexico city, México. June 28th 2016. <http://goo.gl/S6YzX7>

Radio

UNAM estudia el universo con el uso de códigos numéricos

El Universal, Mexico city, México. June 20th 2016. <http://goo.gl/CcGB6z>

Press

Códigos numéricos para la Astrofísica

Agencia informativa CONACYT, Mexico city, México. May 16th 2016.

Press

Academic Service

Committee.....

- Miembro del **Subcomité de Supercomputo** del Comité Técnico de Tecnologías de Información y Comunicación, DGTIC-UNAM, Jan.2026 – now
- Miembro del **Comité de Educación Continua**, Instituto de Astronomía-UNAM, Jan.2026, – now
- Miembro del **Comité de Asuntos Postdoctorales**, Instituto de Astronomía-UNAM, Jan.2026, – now
- **Representante del Tutorado** de Instituto de Astronomía ante el Posgrado en Física-UNAM, Jun 2025, – now.

Organizing Academic Events.....

- **Founder and Chair of the Workshop on Relativistic Astrophysics 2025**, May 8th and 9th, 2025.
- **Founder and Chair of AstroLunch: Astrofísica Relativista**, Instituto de Astronomía-UNAM, Sep. 2024 – now.
- **Chair of the LOC of the EHT Collaboration Meeting summer 2024**, Unidad de Posgrado-UNAM, 2024 May 19th-24th, Ciudad de México.
- **Co-chair of the "Arcadio Poveda's" Colloquium**, IA-UNAM, Sep.2023 – Dec.2025
- **SOC of the First Mexican meeting in route to the Next Generation VLA**, IRyA-UNAM, Michoacán, México, 25th-27th Sep 2023.
- **LOC of the EHT winter collaboration Meeting 2022**, 12th-25th December 2022.
- **Chair of the AstroCoffee seminar**, Institut für Theoretische Physik, Johann Wolfgang Goethe Universität Frankfurt, Oct.2020 – Sep.2022
- **EHT Sgr A* Theory Busy Days (2020-Q4)**, Event Horizon Telescope Collaboration, Nov 30th – Dec 4th 2021.
- **First Student Meeting 2014**, Instituto Tecnológico Superior de los Ríos, November 20th 2015, Tabasco, México.
- **Basic Electronics Conference-Workshop**, Faculty of Science Physics and Mathematics-UMSNH, April 28th & 29th 2006, Morelia, México.
- **1º Regional Meeting of Optic**, September 19th-23th 2005, Morelia, México.

SECIHTI Reviewer.....

- Becas de Posgrado en Ciencias y Humanidades en el Extranjero 2025, SECIHTI⁹.
- Convocatoria 2025 Estancias Posdoctorales por México, SECIHTI.
- Convocatoria Estancias Posdoctorales por México Iniciales 2024, CONAHCYT.

SECIHTI: Secretaría de Ciencia, Humanidades, Tecnología e Innovación

- Convocatoria 2021 Estancias Posdoctorales por México, CONAHCYT.

Journal Reviewer.....

- The European Physical Journal C
- Revista Mexicana de Astronomía y Astrofísica
- Astrophysical Journal
- American Journal of Physics ¹⁰
- Revista Mexicana de Física
- MDPI-Universe ¹¹
- Referee at DiRAC-cluster England: Astronomy, Particle Physics and Nuclear Physics

Computer skills

Languages: Fortran, Python, C/C++

Code Development: CAFE, BHAC, BHOSS

High Performance: OpenMP, MPI, AMR, FMR

Codes: Einstein Toolkit, ehtim

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 Universe 2021, 7(2), 22; <https://doi.org/10.3390/universe7020022>